Fermi–Dirac Statistics in a Semiconductor Crystal

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(Dated: October 22, 2026)

We compute the electronic occupation statistics of a parabolic-band semiconductor by solving the carrier-density integral $n = \int D(E)f(E)\,dE$ numerically for the chemical potential. Writing the integral in terms of the dimensionless Fermi–Dirac integral $F_{1/2}$ keeps the units in one place and makes the integration cutoff a multiple of k_BT. The implementation is validated against both analytic limits: it reproduces the non-degenerate Boltzmann result to 1.3×10^{-4} and the degenerate $T\to0$ Fermi energy to 10^{-7}, and its temperature dependence follows the Sommerfeld expansion to 4×10^{-9}. An interactive Streamlit interface exposes temperature, carrier concentration, effective mass and band gap, and reports E_F, k_F, v_F and T_F alongside a degeneracy flag. We stress a distinction the usual phrasing obscures: the quantity that varies with temperature is the chemical potential $\mu(T)$, not the Fermi energy $E_F=\mu(0)$, and k_F, v_F and T_F are meaningful only when the carrier gas is degenerate — which at room temperature requires $n \gtrsim 10^{18}\,\mathrm{cm}^{-3}$.

I. INTRODUCTION

The electrons in a crystal are indistinguishable spin-$\frac{1}{2}$ fermions, so the Pauli principle limits each state to one of them. Their equilibrium distribution over energy is therefore not the Boltzmann distribution of classical statistics but the Fermi–Dirac distribution. In a metal this distinction is dramatic: the electron gas is degenerate, the occupation is essentially a step, and only electrons within k_BT of the Fermi level participate in transport or heat capacity. In a lightly doped semiconductor the carriers are dilute enough that Fermi–Dirac statistics reduce to Boltzmann statistics and the distinction nearly vanishes.

A semiconductor is interesting precisely because it can be moved between these two regimes by doping. This report computes where the boundary lies, by solving for the chemical potential without approximating the statistics, and uses that to examine how the Fermi level moves with temperature and carrier concentration.

II. THEORY

A. Fermi–Dirac statistics

The mean occupation of a single-particle state of energy E in equilibrium at temperature T is

$$f(E) = \frac{1}{\exp\big((E-\mu)/k_BT\big)+1}, \tag{1}$$

where μ is the chemical potential. Two properties follow immediately and are used throughout. First, $f(\mu)=\frac{1}{2}$ at every temperature, so μ is the energy at which a state is equally likely to be occupied or empty. Second, as $T\to0$, f becomes a step: every state below μ is filled, every state above is empty.

B. Density of states

In a cubic box of side L with periodic boundary conditions the allowed wavevectors form a lattice of spacing $2\pi/L$, so k-space contains $(L/2\pi)^3$ states per unit volume, doubled for spin. Counting states in a shell of radius k and converting to energy with the effective-mass dispersion $E = E_c+\hbar^2k^2/2m^*$ gives the conduction-band density of states per unit volume,

$$D(E) = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}\sqrt{E-E_c}, \qquad E\ge E_c, \tag{2}$$

and zero inside the gap. The square-root edge is the three-dimensional signature: states become available gradually above the band edge rather than all at once.

C. Carrier density fixes the chemical potential

The carrier concentration is the occupied density of states, integrated:

$$n = \int_{E_c}^{\infty} D(E)\,f(E)\,dE. \tag{3}$$

Given n, T and m^*, this is one equation in the single unknown μ. Substituting $x=(E-E_c)/k_BT$ and $\eta=(\mu-E_c)/k_BT$ reduces it to a dimensionless form,

$$n = N_c\,\frac{2}{\sqrt{\pi}}\,F_{1/2}(\eta), \qquad F_{1/2}(\eta)=\int_0^\infty \frac{\sqrt{x}\,dx}{1+e^{\,x-\eta}}, \tag{4}$$

with the effective conduction-band density of states $N_c = 2\left(m^*k_BT/2\pi\hbar^2\right)^{3/2}$. This is the form we solve: it isolates all dimensional factors in N_c and makes the integration cutoff a multiple of k_BT rather than a fixed energy, which matters because the degenerate case extends far above the band edge.

D. The two limits

a. Non-degenerate ($\eta \ll -1$). The $+1$ in the denominator of (1) is negligible, $F_{1/2}(\eta) \to (\sqrt{\pi}/2)e^{\eta}$, and (4) collapses to the Boltzmann result

$$n = N_c \exp\left(\frac{\mu - E_c}{k_B T}\right). \tag{5}$$

The chemical potential lies *below* the band edge, there is no Fermi surface, and the carriers behave as a classical gas.

b. Degenerate ($\eta \gg 1$). Here $F_{1/2}(\eta) \to \frac{2}{3}\eta^{3/2}$ and (4) inverts to

$$E_F \equiv \mu(T{=}0) - E_c = \frac{\hbar^2}{2m^*}\left(3\pi^2 n\right)^{2/3}, \tag{6}$$

the familiar free-electron-gas result. The corresponding Fermi wavevector, velocity and temperature are

$$k_F = (3\pi^2 n)^{1/3}, \quad v_F = \frac{\hbar k_F}{m^*}, \quad T_F = \frac{E_F}{k_B}. \tag{7}$$

E. Two points of care

a. E_F is not $\mu(T)$. The two coincide only at $T = 0$. Equation (6) defines E_F as a function of n alone, with no temperature in it; the quantity that moves when the temperature changes is $\mu(T)$, obtained from (4). Where this report plots a "Fermi level versus temperature" it plots $\mu(T)$, and the distinction is not cosmetic: in Fig. 5(a) the $n = 10^{16}\,\mathrm{cm}^{-3}$ curve falls by 0.3 eV over the range shown while its E_F is fixed at 2.5 meV.

b. k_F, v_F and T_F presuppose degeneracy. They describe a filled Fermi sphere. For $n = 10^{16}\,\mathrm{cm}^{-3}$ in GaAs, Eq. (7) returns $T_F = 29$ K — far below room temperature, which is precisely the statement that the gas is *not* degenerate at 300 K and that these quantities are formal numbers rather than physical ones. The interface therefore reports them as asked and displays η beside them so the regime is never in doubt.

III. NUMERICAL METHOD

1. Evaluate $F_{1/2}(\eta)$ by quadrature, with the upper limit at $\max(\eta, 0) + 60$; the integrand falls off as $e^{-(x-\eta)}$, so this is far past negligible.
2. Solve $n(\eta) = n_{\mathrm{target}}$ for η with Brent's method.
3. Bracket by widening geometrically until the residual changes sign. A fixed bracket fails at low temperature, where $n(\eta)$ is nearly a step.
4. Return $\mu = E_c + \eta k_B T$.

Evaluating $f(E)$ through the logistic function `expit` avoids overflow when $|E - \mu| \gg k_B T$, which otherwise produces `inf` and a silent `nan`.

A. Validation

Six checks run in `test_fermi.py`; all pass.

Check	Result
Non-degenerate limit vs Eq. (5)	1.3×10^{-4}
Degenerate $T \to 0$ limit vs Eq. (6)	9.6×10^{-8}
Carrier-density round trip	4.9×10^{-14}
$f(\mu) = \frac{1}{2}$	exact
Sommerfeld expansion for $\mu(T)$	3.6×10^{-9}
$D(E)$ zero in the gap, analytic above E_c	exact

TABLE I. Validation results, as maximum relative errors.

The fifth is the strongest: in the degenerate regime the Sommerfeld expansion predicts

$$\mu(T) \simeq E_F\left[1 - \frac{\pi^2}{12}\left(\frac{k_B T}{E_F}\right)^2\right], \tag{8}$$

and the numerical solution follows this T^2 law to nine digits, as Fig. 5(b) shows. That tests the temperature dependence quantitatively rather than merely checking its sign.

IV. THE INTERACTIVE INTERFACE

The interface is written with Streamlit and launched with `streamlit run app.py`. The sidebar takes the band gap, effective mass, temperature, carrier concentration and energy range, with presets for GaAs, Si and Ge. The main panel reports $\mu - E_c$, E_F, k_F, v_F and T_F, states which regime the parameters fall in, and plots the band edges with $f(E)$ on a shared energy axis, $D(E)$ with its occupied part, and both sweeps. Every change recomputes the full integral; nothing is approximated for speed.

V. RESULTS

Unless stated otherwise the material is GaAs-like, $E_g = 1.42$ eV and $m^* = 0.067\,m_e$, at $T = 300$ K.

A. The three regimes

n (cm^{-3})	$\mu - E_c$ (meV)	η	E_F (meV)	T_F (K)
10^{16}	−97.3	−3.77	2.5	29
10^{18}	41.9	1.62	54.4	632
10^{19}	250.4	9.69	252.6	2931

TABLE II. GaAs at 300 K across three doping levels. Degeneracy requires $T \ll T_F$, which the last column makes explicit.

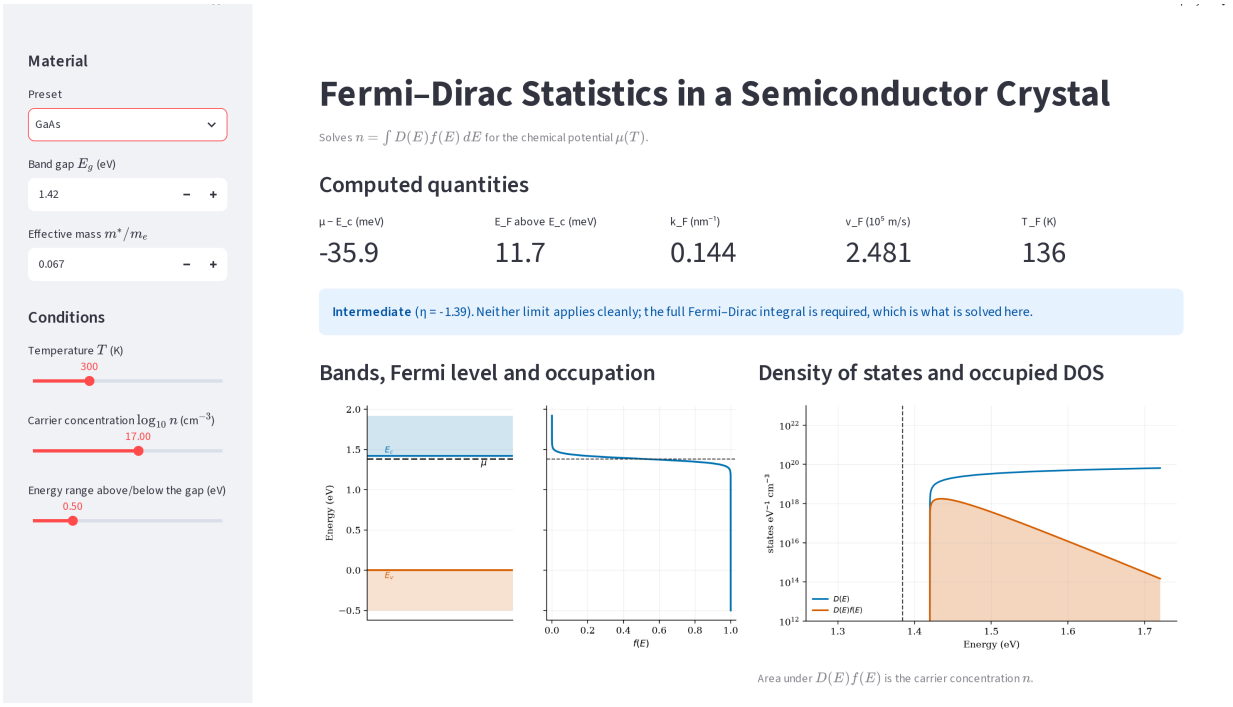


FIG. 1. The interface on the GaAs preset at $T = 300$ K and $n = 10^{17}\,\mathrm{cm}^{-3}$. The banner reports $\eta = -1.39$: neither limit applies cleanly here, which is why the full integral is solved rather than either closed form.

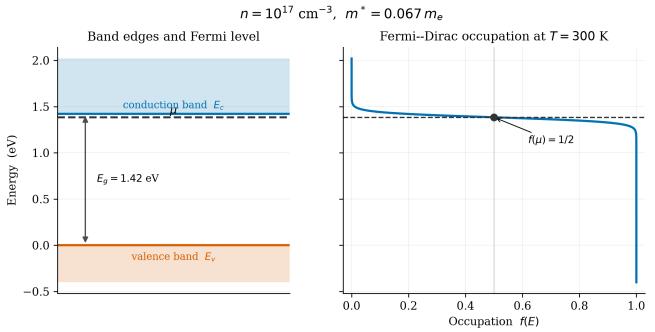


FIG. 2. Band edges, chemical potential and the Fermi–Dirac occupation on a shared energy axis, at $n = 10^{17}\,\mathrm{cm}^{-3}$. $f(E)$ passes through $\frac{1}{2}$ exactly at μ.

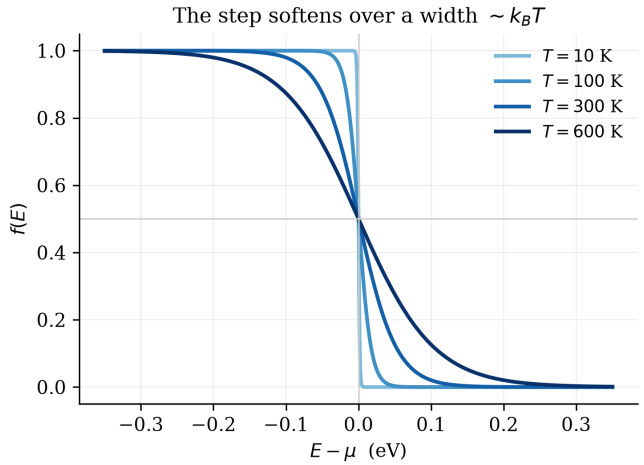


FIG. 3. The occupation step softens over a width of order $k_B T$. At 10 K it is nearly the $T = 0$ step; at 600 K it is smeared over more than 0.2 eV.

Table II locates the crossover. At $10^{16}\,\mathrm{cm}^{-3}$ the chemical potential sits 97 meV *below* the band edge and $T_F = 29$ K is a tenth of the actual temperature: the gas is classical. At $10^{19}\,\mathrm{cm}^{-3}$, μ lies 250 meV *above* the edge and $T_F = 2931$ K, so 300 K is effectively a low temperature and μ has risen to within 1% of E_F. The case at $10^{18}\,\mathrm{cm}^{-3}$ sits between them, with $\eta = 1.62$: neither closed form is reliable there and the full integral earns its place.

a. Why the doping stops at 10^{19} cm^{-3}. It is tempting to push further to make the degenerate limit cleaner, but the model would stop describing the material. At $10^{20}\,\mathrm{cm}^{-3}$ Eq. (6) puts E_F at 1.17 eV above E_c, which is $0.83E_g$ and lies far above the L and X satellite valleys of GaAs at roughly 0.29 and 0.48 eV. Those valleys would hold most of the carriers, so a single parabolic Γ valley is no longer the band structure and the computed E_F, while arithmetically correct, would be physically meaningless. Doping that heavy is also beyond what n-type GaAs achieves in practice, since Si dopants become amphoteric above roughly $10^{19}\,\mathrm{cm}^{-3}$. We therefore keep every reported case within the range where the stated model holds.

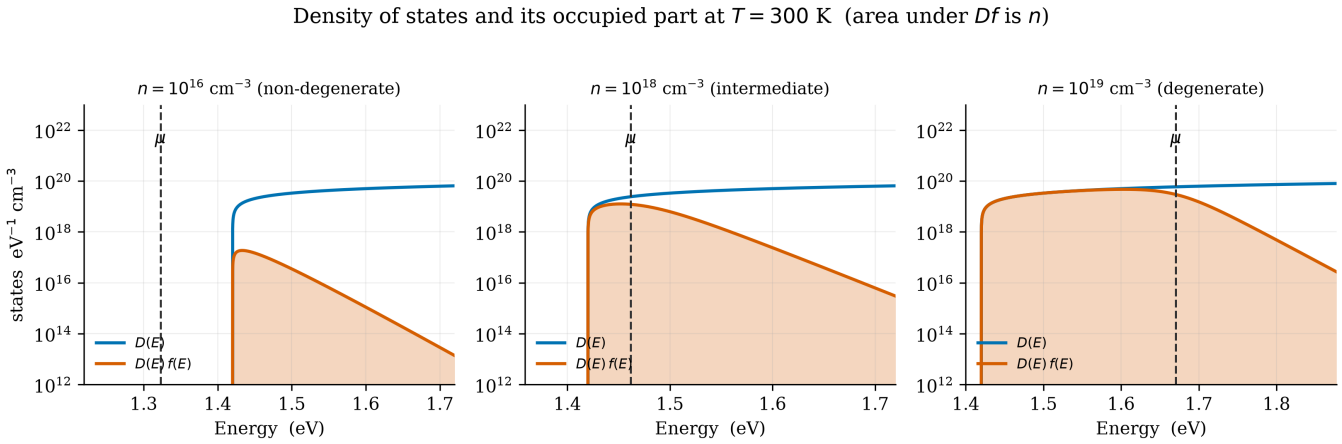


FIG. 4. $D(E)$ and the occupied density of states $D(E)f(E)$ across the three regimes, at 300 K. A logarithmic axis is necessary: at $10^{16}\,\mathrm{cm}^{-3}$ the occupied curve lies some four orders of magnitude below $D(E)$ and would be indistinguishable from zero on a linear scale. The area under $D(E)f(E)$ is n in every panel.

Figure 4 shows the same crossover in the occupation itself. In the non-degenerate panel $D(E)f(E)$ falls exponentially above the band edge — a straight line on the logarithmic axis, the Boltzmann tail of Eq. (5). In the degenerate panel it instead tracks $D(E)$ itself right up to μ and then drops within a few k_BT: the states are simply filled to the chemical potential, as they would be in a metal.

B. Fermi level versus temperature

Figure 5(a) shows μ falling as temperature rises, and the reason is the shape of $D(E)$. The density of states grows as $\sqrt{E - E_c}$, so there are more states above μ than below it within the thermal window. Raising T promotes carriers upward into that larger reservoir, and μ must fall to keep the total fixed. At low doping it falls below the band edge entirely, which is the formal statement that the carriers have gone classical.

The degenerate case is far flatter, and Fig. 5(b) shows why: the shift is the T^2 Sommerfeld correction of Eq. (8), which stays small while $k_BT \ll E_F$. The numerical points follow the analytic curve closely up to about 300 K and then begin to separate, reaching a 0.5% discrepancy at 700 K where $k_BT/E_F \approx 0.24$. That departure is the expected behaviour of an asymptotic expansion, and seeing where it begins is more informative than a case chosen to hide it.

C. Fermi level versus carrier concentration

Figure 6 shows μ rising with doping, logarithmically at first and then more steeply. The logarithmic stretch is Eq. (5) rearranged, $\mu = E_c + k_BT\ln(n/N_c)$: in the non-degenerate regime the chemical potential depends only on the *ratio* n/N_c. The curve crosses E_c near $n \simeq N_c$, and beyond that the Pauli principle takes over — further carriers must occupy higher states, and μ climbs as $n^{2/3}$ from Eq. (6), far faster than logarithmically.

Raising the temperature shifts the crossing to higher n, because $N_c \propto T^{3/2}$: a hot semiconductor has more thermally accessible states and needs heavier doping before degeneracy sets in.

D. Different materials

Material	E_g (eV)	m^*/m_e	$\mu - E_c$ (meV)	η
GaAs	1.42	0.067	−35.9	−1.39
Si	1.12	0.260	−90.3	−3.49
Ge	0.66	0.120	−59.7	−2.31

TABLE III. The three presets at $n = 10^{17}\,\mathrm{cm}^{-3}$ and 300 K.

Table III isolates the role of the effective mass. All three materials carry the same carrier concentration, yet GaAs is closest to degeneracy and silicon furthest from it. The band gap is irrelevant to this — it only sets where E_c sits — and the ordering follows m^* instead, through $N_c \propto (m^*)^{3/2}$. A light effective mass means few available states, so a given carrier density fills them further up and pushes μ toward the band edge. This is why GaAs, with the lightest electrons of the three, becomes degenerate at the lowest doping.

VI. CONCLUSIONS

Solving the carrier-density integral directly, rather than assuming either limit, gives a method that spans the classical and degenerate regimes with one code path

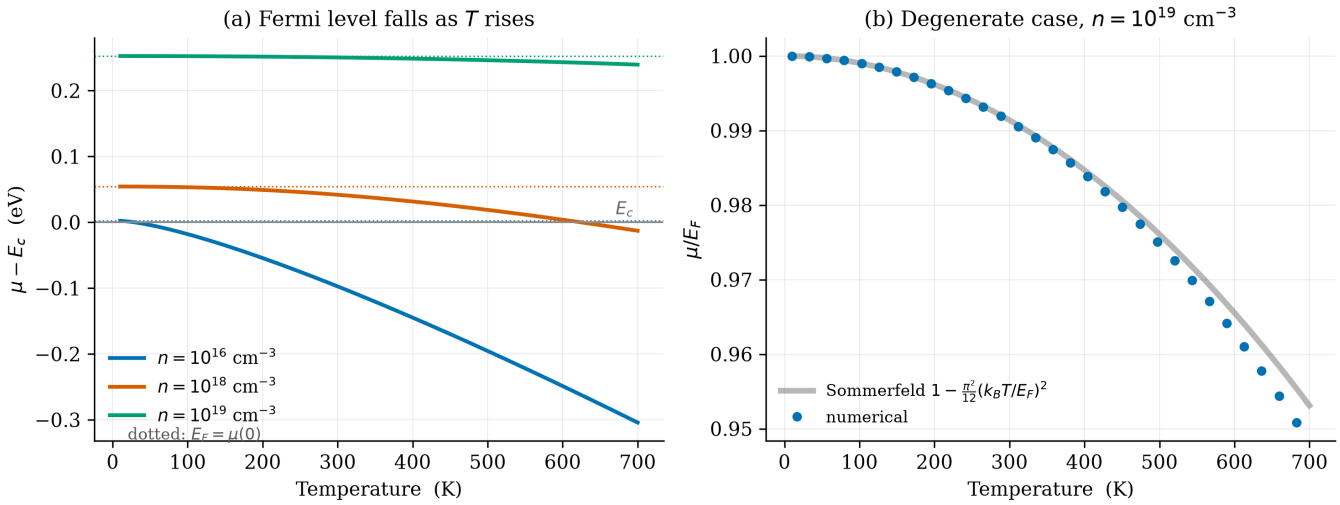


FIG. 5. (a) $\mu(T)$ at three carrier concentrations; dotted lines mark $E_F = \mu(0)$. (b) The degenerate case at $10^{19}\,\mathrm{cm}^{-3}$ against the Sommerfeld expansion Eq. (8). The numerical points follow the analytic curve, departing from it by 0.5% only by 700 K, where k_BT/E_F is no longer small and the asymptotic expansion begins to fail as expected.

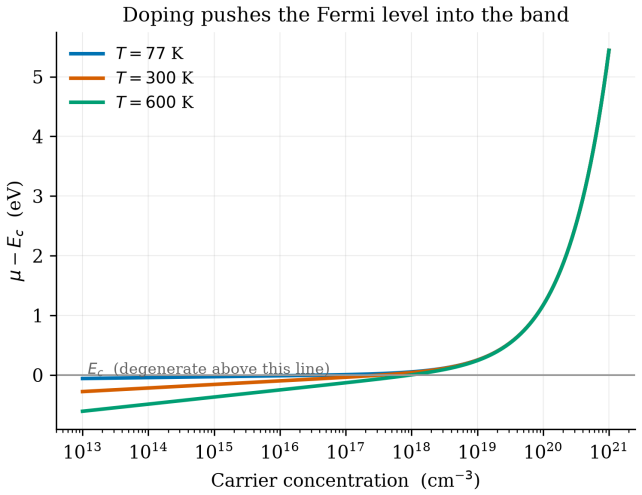


FIG. 6. $\mu - E_c$ against carrier concentration at three temperatures. The curve crosses into the band near $n \sim N_c$, which is $4.35 \times 10^{17}\,\mathrm{cm}^{-3}$ for GaAs at 300 K.

and reproduces both closed forms where they apply — to 1.3×10^{-4} for the Boltzmann limit and 10^{-7} for the degenerate limit. Applying it:

- The classical–degenerate boundary sits near $n \sim N_c$, which for GaAs at 300 K is $4.35 \times 10^{17}\,\mathrm{cm}^{-3}$. Equivalently, degeneracy requires $T \ll T_F$.
- μ falls with temperature because $D(E) \propto \sqrt{E - E_c}$ offers more states above the chemical potential than below it; at low doping it drops below the band edge.
- In the degenerate regime that fall is the T^2 Sommerfeld correction, which the numerics reproduce until k_BT/E_F ceases to be small.
- μ rises logarithmically with n while non-degenerate and as $n^{2/3}$ once degenerate.
- Effective mass, not band gap, controls how readily a material becomes degenerate.

The model's limitations bound where it may be used. The band is a single parabolic valley: real silicon has six equivalent X valleys, which raise N_c by that factor, and GaAs has L and X valleys only 0.29 and 0.48 eV above Γ, so the model fails once E_F approaches them — the reason the doping range above stops at $10^{19}\,\mathrm{cm}^{-3}$. Band non-parabolicity would reduce E_F further at high doping. We also neglect the valence band and dopant ionisation statistics, and treat n as an input rather than solving for it self-consistently from the dopant levels.

REPRODUCING THIS WORK

```
uv venv --python 3.12 .venv
uv pip install -r requirements.txt
cd question_5
python test_fermi.py      # checks
python make_figures.py    # figures
streamlit run app.py      # interface
```

Appendix A: Complete source code

Listings are included directly from the source files, so they cannot drift from the code that produced the results above.

1. `fermi.py` — physics core

```python
"""Fermi-Dirac statistics for a parabolic-band semiconductor.

Single conduction band in the effective-mass approximation, 3D. Energies are
handled in joules internally; 'EV' converts to and from electronvolts.

Carrier density is evaluated through the dimensionless Fermi-Dirac integral

    n = N_c (2/sqrt(pi)) F_{1/2}(eta),   eta = (mu - E_c) / kT

which keeps the units in one place and makes the integration cutoff a multiple
of kT rather than a fixed energy.
"""

import numpy as np
from scipy import integrate, optimize
from scipy.special import expit

# ---- constants (SI) ----
HBAR = 1.054571817e-34
M_E = 9.1093837015e-31
K_B = 1.380649e-23
EV = 1.602176634e-19          # joules per electronvolt


# ---- occupation ----

def fermi_dirac(E, mu, T):
    """f(E) = 1/(exp((E-mu)/kT) + 1), written to not overflow."""
    # expit(x) = 1/(1+exp(-x)), so f = expit(-(E-mu)/kT) with no large exp
    return expit(-(np.asarray(E, dtype=float) - mu) / (K_B * T))


# ---- density of states ----

def dos(E, m_eff, Ec=0.0):
    """Conduction-band DOS per unit volume per unit energy; zero below Ec."""
    E = np.asarray(E, dtype=float)
    out = np.zeros_like(E)
    above = E > Ec
    out[above] = (1.0 / (2 * np.pi**2)) * (2 * m_eff / HBAR**2) ** 1.5 \
        * np.sqrt(E[above] - Ec)
    return out


def n_c(m_eff, T):
    """Effective conduction-band density of states N_c."""
    return 2.0 * (m_eff * K_B * T / (2 * np.pi * HBAR**2)) ** 1.5


# ---- carrier density ----

def fermi_integral_half(eta):
    """F_{1/2}(eta) = int_0^inf sqrt(x)/(1+exp(x-eta)) dx."""
    # integrand decays like exp(-(x-eta)), so eta + 60 is far past negligible
    upper = max(eta, 0.0) + 60.0
    val, _ = integrate.quad(lambda x: np.sqrt(x) * expit(eta - x), 0.0, upper,
                            limit=200)
    return val


def carrier_density(mu, T, m_eff, Ec=0.0):
    """n = int D(E) f(E) dE, via the dimensionless integral."""
    eta = (mu - Ec) / (K_B * T)
    return n_c(m_eff, T) * (2.0 / np.sqrt(np.pi)) * fermi_integral_half(eta)


def solve_mu(n, T, m_eff, Ec=0.0):
    """Chemical potential mu(T) from charge neutrality, by root-finding on eta.

    At low T the integral is nearly a step in eta, so the bracket is widened
    geometrically until it straddles the root instead of being fixed.
```

```
    """
    target = float(n)

    def residual(eta):
        return n_c(m_eff, T) * (2.0 / np.sqrt(np.pi)) * fermi_integral_half(eta) - target

    lo, hi = -5.0, 5.0
    for _ in range(80):
        if residual(lo) < 0 < residual(hi):
            break
        lo *= 2.0 if residual(lo) >= 0 else 1.0
        hi *= 2.0 if residual(hi) <= 0 else 1.0
        if lo > -1e-6:
            lo = -5.0
        if abs(lo) > 1e12 or abs(hi) > 1e12:
            raise RuntimeError("could not bracket the chemical potential")
    else:
        raise RuntimeError("could not bracket the chemical potential")

    eta = optimize.brentq(residual, lo, hi, xtol=1e-12, rtol=1e-14)
    return Ec + eta * K_B * T


def boltzmann_mu(n, T, m_eff, Ec=0.0):
    """Non-degenerate closed form: n = N_c exp((mu-Ec)/kT). For validation."""
    return Ec + K_B * T * np.log(n / n_c(m_eff, T))


# ---- degenerate-gas quantities ----

def fermi_quantities(n, m_eff):
    """E_F (above Ec), k_F, v_F, T_F for a filled Fermi sphere of density n.

    These describe a degenerate gas. For a non-degenerate semiconductor they
    are formal numbers; 'degeneracy' below says which regime applies.
    """
    k_F = (3 * np.pi**2 * n) ** (1 / 3)
    E_F = HBAR**2 * k_F**2 / (2 * m_eff)
    return {"E_F": E_F, "k_F": k_F, "v_F": HBAR * k_F / m_eff, "T_F": E_F / K_B}


def degeneracy(mu, T, Ec=0.0):
    """eta = (mu - Ec)/kT.  eta >> 1 degenerate, eta << -1 non-degenerate."""
    return (mu - Ec) / (K_B * T)


def sommerfeld_mu(n, T, m_eff, Ec=0.0):
    """Degenerate low-T expansion: mu ~ E_F[1 - (pi^2/12)(kT/E_F)^2]."""
    E_F = fermi_quantities(n, m_eff)["E_F"]
    return Ec + E_F * (1.0 - (np.pi**2 / 12.0) * (K_B * T / E_F) ** 2)
```

2. test_fermi.py — validation suite

```
"""Validation checks from PLAN.md. Run: python test_fermi.py"""

import numpy as np
import fermi as f

M = 0.067 * f.M_E          # GaAs-like effective mass
CM3 = 1e6                  # cm^-3 -> m^-3


def check(name, ok, detail=""):
    print(f"[{'PASS' if ok else 'FAIL'}] {name}" + (f"  {detail}" if detail else ""))
    return ok


# ---- 1. non-degenerate limit matches the Boltzmann closed form ----

def test_boltzmann():
    worst = 0.0
    for n_cm3 in (1e12, 1e14, 1e15):
        n = n_cm3 * CM3
        num = f.solve_mu(n, 300.0, M)
        ana = f.boltzmann_mu(n, 300.0, M)
        worst = max(worst, abs(num - ana) / abs(ana))
    return check("non-degenerate vs Boltzmann", worst < 2e-3, f"max rel err {worst:.2e}")


```

```python
# ---- 2. degenerate limit: mu -> E_F as T -> 0 ----

def test_degenerate():
    n = 1e19 * CM3
    E_F = f.fermi_quantities(n, M)["E_F"]
    mu = f.solve_mu(n, 1.0, M)          # 1 K is deep in the degenerate regime
    err = abs(mu - E_F) / E_F
    return check("degenerate T->0 vs E_F", err < 1e-4, f"rel err {err:.2e}")


# ---- 3. round trip: the solved mu reproduces n ----

def test_round_trip():
    worst = 0.0
    for n_cm3 in (1e14, 1e16, 1e18, 1e20):
        for T in (50.0, 300.0, 600.0):
            n = n_cm3 * CM3
            back = f.carrier_density(f.solve_mu(n, T, M), T, M)
            worst = max(worst, abs(back - n) / n)
    return check("carrier-density round trip", worst < 1e-8, f"max rel err {worst:.2e}")


# ---- 4. f(mu) = 1/2 at any temperature ----

def test_half_occupancy():
    worst = max(abs(f.fermi_dirac(0.3 * f.EV, 0.3 * f.EV, T) - 0.5)
                for T in (1.0, 77.0, 300.0, 1000.0))
    return check("f(mu) = 1/2", worst < 1e-15, f"max dev {worst:.2e}")


# ---- 5. Sommerfeld expansion in the degenerate regime ----

def test_sommerfeld():
    n = 1e20 * CM3                      # strongly degenerate, E_F ~ 1 eV
    worst = 0.0
    for T in (20.0, 50.0, 100.0):
        num = f.solve_mu(n, T, M)
        som = f.sommerfeld_mu(n, T, M)
        worst = max(worst, abs(num - som) / abs(som))
    return check("degenerate mu(T) vs Sommerfeld", worst < 1e-4,
                 f"max rel err {worst:.2e}")


# ---- 6. DOS is zero in the gap and matches the analytic form above Ec ----

def test_dos():
    Ec = 1.42 * f.EV
    below = f.dos(np.linspace(0.0, Ec, 50), M, Ec)
    E = Ec + np.array([0.01, 0.1, 0.5]) * f.EV
    ana = (1 / (2 * np.pi**2)) * (2 * M / f.HBAR**2) ** 1.5 * np.sqrt(E - Ec)
    return check("DOS zero in the gap", np.all(below == 0)) and \
        check("DOS matches analytic form",
              np.allclose(f.dos(E, M, Ec), ana, rtol=1e-12))


if __name__ == "__main__":
    results = [test_boltzmann(), test_degenerate(), test_round_trip(),
               test_half_occupancy(), test_sommerfeld(), test_dos()]
    print(f"\n{sum(results)}/{len(results)} checks passed")
    raise SystemExit(0 if all(results) else 1)
```

3. make_figures.py — figure generation

```python
"""Regenerates every figure in figures/. Run: python make_figures.py"""

import os
import sys

import numpy as np
import matplotlib.pyplot as plt

sys.path.insert(0, os.path.join(os.path.dirname(__file__), "..", "common"))
import style
import fermi as f

FIG = os.path.join(os.path.dirname(__file__), "figures")

# GaAs-like reference material, energies measured from the valence-band edge
EG = 1.42 * f.EV
```

```
M = 0.067 * f.M_E
CM3 = 1e6
N_REF = 1e17 * CM3
T_REF = 300.0


def ev(E):
    return E / f.EV


# ---- band diagram with f(E) alongside ----

def fig_bands():
    mu = f.solve_mu(N_REF, T_REF, M, EG)
    E = np.linspace(-0.4 * f.EV, EG + 0.6 * f.EV, 2000)

    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(9.5, 4.0), sharey=True,
                                   gridspec_kw=dict(width_ratios=[1.0, 1.3]))

    # left: the bands themselves
    ax1.axhspan(ev(EG), ev(E[-1]), color=style.ACOUSTIC, alpha=0.18, lw=0)
    ax1.axhspan(ev(E[0]), 0.0, color=style.OPTICAL[0], alpha=0.18, lw=0)
    ax1.axhline(ev(EG), color=style.ACOUSTIC, lw=2)
    ax1.axhline(0.0, color=style.OPTICAL[0], lw=2)
    ax1.axhline(ev(mu), color=style.EXACT, lw=1.8, ls="--")
    ax1.text(0.5, ev(EG) + 0.06, r"conduction band  $E_c$", ha="center", fontsize=9,
             color=style.ACOUSTIC)
    ax1.text(0.5, -0.16, r"valence band  $E_v$", ha="center", fontsize=9,
             color=style.OPTICAL[0])
    ax1.text(0.5, ev(mu) + 0.04, r"$\mu$", ha="center", fontsize=10)
    ax1.annotate("", xy=(0.12, ev(EG)), xytext=(0.12, 0.0),
                 arrowprops=dict(arrowstyle="<|-|>", color="0.3", lw=1.2))
    ax1.text(0.17, ev(EG) / 2, rf"$E_g = {EG/f.EV:.2f}$ eV", fontsize=9, va="center")
    ax1.set_xticks([])
    ax1.set_ylabel("Energy  (eV)")
    ax1.set_title("Band edges and Fermi level")

    # right: occupation on the same energy axis
    ax2.plot(f.fermi_dirac(E, mu, T_REF), ev(E), color=style.ACOUSTIC, lw=2)
    ax2.axhline(ev(mu), color=style.EXACT, lw=1.2, ls="--")
    ax2.axvline(0.5, color="0.8", lw=0.8)
    ax2.plot([0.5], [ev(mu)], marker="o", ms=6, color=style.EXACT, zorder=4)
    ax2.annotate(r"$f(\mu)=1/2$", xy=(0.5, ev(mu)), xytext=(0.62, ev(mu) - 0.28),
                 fontsize=9, arrowprops=dict(arrowstyle="->", lw=0.8))
    ax2.set_xlabel(r"Occupation  $f(E)$")
    ax2.set_xlim(-0.03, 1.03)
    ax2.set_title(rf"Fermi--Dirac occupation at $T={T_REF:.0f}$ K")

    fig.suptitle(rf"$n = 10^{{17}}$ cm$^{{-3}}$,  $m^*={M/f.M_E:.3f}\,m_e$", y=1.01)
    fig.savefig(f"{FIG}/bands.png")
    plt.close(fig)


# ---- f(E) sharpening as T falls ----

def fig_occupation():
    mu = f.solve_mu(1e19 * CM3, T_REF, M, EG)
    E = np.linspace(mu - 0.35 * f.EV, mu + 0.35 * f.EV, 2000)

    fig, ax = plt.subplots(figsize=(5.4, 3.6))
    cols = plt.cm.Blues(np.linspace(0.45, 1.0, 4))
    for T, c in zip([10.0, 100.0, 300.0, 600.0], cols):
        ax.plot(ev(E - mu), f.fermi_dirac(E, mu, T), color=c, label=f"$T={T:.0f}$ K")
    ax.axvline(0.0, color="0.8", lw=0.8)
    ax.axhline(0.5, color="0.8", lw=0.8)
    ax.set_xlabel(r"$E-\mu$  (eV)")
    ax.set_ylabel(r"$f(E)$")
    ax.set_title("The step softens over a width $\\sim k_BT$")
    ax.legend()
    fig.savefig(f"{FIG}/occupation.png")
    plt.close(fig)


# ---- D(E), f(E) and the occupied DOS ----

def dos_per_ev_cm3(E, m_eff, Ec):
    """DOS in the conventional semiconductor units: states eV^-1 cm^-3."""
    return f.dos(E, m_eff, Ec) * f.EV * 1e-6


def fig_dos():
    """D(E) and D(E)f(E) across three doping regimes.

```

```
    Log y: at n = 1e16 the occupied fraction is ~1e-7 of D(E), so on a linear
    axis the occupied curve would be a flat zero and show nothing.
    """
    fig, axes = plt.subplots(1, 3, figsize=(11, 3.5))

    for ax, (n_cm3, lab) in zip(axes, [(1e16, "non-degenerate"),
                                       (1e18, "intermediate"),
                                       (1e19, "degenerate")]):
        n = n_cm3 * CM3
        mu = f.solve_mu(n, T_REF, M, EG)
        # window follows mu, so the occupied region is bounded by physics
        # window must contain mu, which lies BELOW Ec when non-degenerate
        lo = min(EG - 0.02 * f.EV, mu - 4 * f.K_B * T_REF)
        hi = max(EG + 0.3 * f.EV, mu + 8 * f.K_B * T_REF)
        E = np.linspace(lo, hi, 3000)

        D = dos_per_ev_cm3(E, M, EG)
        occ = D * f.fermi_dirac(E, mu, T_REF)

        ax.semilogy(ev(E), D, color=style.ACOUSTIC, lw=1.8, label=r"$D(E)$")
        ax.semilogy(ev(E), np.maximum(occ, 1e10), color=style.OPTICAL[0], lw=1.8,
                    label=r"$D(E)\,f(E)$")
        ax.fill_between(ev(E), np.maximum(occ, 1e10), 1e10,
                        color=style.OPTICAL[0], alpha=0.25, lw=0)
        ax.axvline(ev(mu), color=style.EXACT, ls="--", lw=1.2)
        ax.annotate(r"$\mu$", xy=(ev(mu), 3e21), fontsize=10, ha="center")
        ax.set_ylim(1e12, 1e23)
        ax.set_xlim(ev(E[0]), ev(E[-1]))
        ax.set_xlabel("Energy  (eV)")
        ax.set_title(rf"$n=10^{{{int(np.log10(n_cm3))}}}$ cm$^{{-3}}$ ({lab})",
                     fontsize=9.5)
        ax.legend(fontsize=8, loc="lower left")

    axes[0].set_ylabel(r"states  eV$^{-1}$ cm$^{-3}$")
    fig.suptitle(rf"Density of states and its occupied part at $T={T_REF:.0f}$ K"
                 r"  (area under $D f$ is $n$)", y=1.03)
    fig.tight_layout()
    fig.savefig(f"{FIG}/dos_occupied.png")
    plt.close(fig)


# ---- mu versus temperature ----

def fig_mu_vs_T():
    """Two panels: the crossover below E_c, and the degenerate T^2 law."""
    T = np.linspace(10.0, 700.0, 160)
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 3.9))

    # (a) moderate doping -- mu drops below the band edge as T rises
    for n_cm3, c in zip([1e16, 1e18, 1e19],
                        [style.ACOUSTIC, style.OPTICAL[0], style.OPTICAL[1]]):
        n = n_cm3 * CM3
        mu = np.array([f.solve_mu(n, t, M, EG) for t in T])
        ax1.plot(T, ev(mu - EG), color=c,
                 label=rf"$n=10^{{{int(np.log10(n_cm3))}}}$ cm$^{{-3}}$")
        ax1.axhline(ev(f.fermi_quantities(n, M)["E_F"]), color=c, lw=0.8, ls=":")

    ax1.axhline(0.0, color="0.55", lw=1.0)
    ax1.text(695, 0.012, r"$E_c$", ha="right", fontsize=9, color="0.4")
    ax1.text(20, -0.33, r"dotted: $E_F=\mu(0)$", fontsize=8.5, color="0.35")
    ax1.set_xlabel("Temperature  (K)")
    ax1.set_ylabel(r"$\mu - E_c$  (eV)")
    ax1.set_title("(a) Fermi level falls as $T$ rises")
    ax1.legend(loc="lower left", fontsize=9)

    # (b) strongly degenerate -- the numerical mu follows the Sommerfeld law
    n = 1e19 * CM3
    E_F = f.fermi_quantities(n, M)["E_F"]
    Td = np.linspace(10.0, 700.0, 120)
    mu_num = np.array([f.solve_mu(n, t, M, EG) for t in Td])
    mu_som = np.array([f.sommerfeld_mu(n, t, M, EG) for t in Td])

    ax2.plot(Td, (mu_som - EG) / E_F, color=style.EXACT, lw=3.0, alpha=0.35,
             label=r"Sommerfeld $1-\frac{\pi^2}{12}(k_BT/E_F)^2$")
    ax2.plot(Td[::4], (mu_num[::4] - EG) / E_F, ls="none", marker="o", ms=4,
             color=style.ACOUSTIC, label="numerical")
    ax2.set_xlabel("Temperature  (K)")
    ax2.set_ylabel(r"$\mu / E_F$")
    ax2.set_title(rf"(b) Degenerate case, $n=10^{{19}}$ cm$^{{-3}}$")
    ax2.legend(loc="lower left", fontsize=8.5)

    fig.tight_layout()
    fig.savefig(f"{FIG}/mu_vs_T.png")
```

```
    plt.close(fig)


# ---- mu versus carrier concentration ----

def fig_mu_vs_n():
    n_cm3 = np.logspace(13, 21, 200)
    fig, ax = plt.subplots(figsize=(5.8, 4.0))

    for T, c in zip([77.0, 300.0, 600.0], [style.ACOUSTIC, style.OPTICAL[0],
                                           style.OPTICAL[1]]):
        mu = np.array([f.solve_mu(x * CM3, T, M, EG) for x in n_cm3])
        ax.semilogx(n_cm3, ev(mu - EG), color=c, label=f"$T={T:.0f}$ K")

    ax.axhline(0.0, color="0.6", lw=1.0)
    ax.text(1.2e13, 0.02, r"$E_c$  (degenerate above this line)", fontsize=8.5,
            color="0.4")
    ax.set_xlabel(r"Carrier concentration  (cm$^{-3}$)")
    ax.set_ylabel(r"$\mu - E_c$  (eV)")
    ax.set_title("Doping pushes the Fermi level into the band")
    ax.legend(loc="upper left", fontsize=9)
    fig.savefig(f"{FIG}/mu_vs_n.png")
    plt.close(fig)


if __name__ == "__main__":
    style.apply()
    os.makedirs(FIG, exist_ok=True)
    fig_bands()
    fig_occupation()
    fig_dos()
    fig_mu_vs_T()
    fig_mu_vs_n()
    print("figures written to", os.path.abspath(FIG))
```

4. app.py — interactive interface

```
"""Streamlit interface for Fermi-Dirac statistics in a semiconductor.

Run:  streamlit run app.py
"""

import os
import sys

import numpy as np
import pandas as pd
import streamlit as st
import matplotlib.pyplot as plt

sys.path.insert(0, os.path.join(os.path.dirname(__file__), "..", "common"))
import style
import fermi as f

style.apply()
st.set_page_config(page_title="Fermi\u2013Dirac Statistics in a Semiconductor",
                   layout="wide")
st.title("Fermi\u2013Dirac Statistics in a Semiconductor Crystal")
st.caption(r"Solves $n=\int D(E)f(E)\,dE$ for the chemical potential $\mu(T)$.")

CM3 = 1e6


def ev(E):
    return E / f.EV


# ---- inputs ----

with st.sidebar:
    st.header("Material")
    preset = st.selectbox("Preset", ["Custom", "GaAs", "Si", "Ge"])
    mats = {"GaAs": (1.42, 0.067), "Si": (1.12, 0.26), "Ge": (0.66, 0.12)}
    if preset in mats:
        eg_d, me_d = mats[preset]
    else:
        eg_d, me_d = 1.42, 0.067

    Eg = st.number_input("Band gap $E_g$ (eV)", 0.1, 6.0, float(eg_d), 0.01,
```

```
                        key=f"eg_{preset}")
    m_rel = st.number_input(r"Effective mass $m^*/m_e$", 0.01, 5.0, float(me_d),
                            0.001, format="%.3f", key=f"me_{preset}")

    st.header("Conditions")
    T = st.slider("Temperature $T$ (K)", 1, 1000, 300)
    log_n = st.slider(r"Carrier concentration  $\log_{10} n$ (cm$^{-3}$)",
                      12.0, 21.0, 17.0, 0.1)
    span = st.slider("Energy range above/below the gap (eV)", 0.1, 2.0, 0.5, 0.1)

n = 10.0 ** log_n * CM3
m_eff = m_rel * f.M_E
Ec = Eg * f.EV

# ---- solve ----

mu = f.solve_mu(n, T, m_eff, Ec)
eta = f.degeneracy(mu, T, Ec)
q = f.fermi_quantities(n, m_eff)

# ---- computed quantities ----

st.subheader("Computed quantities")
c1, c2, c3, c4, c5 = st.columns(5)
c1.metric("μ − E_c  (meV)", f"{ev(mu - Ec)*1000:.1f}")
c2.metric("E_F above E_c  (meV)", f"{ev(q['E_F'])*1000:.1f}")
c3.metric("k_F  (nm$^{-1}$)", f"{q['k_F']/1e9:.3f}")
c4.metric("v_F  ($10^5$ m/s)", f"{q['v_F']/1e5:.3f}")
c5.metric("T_F  (K)", f"{q['T_F']:.0f}")

if eta > 2:
    st.success(f"**Degenerate** (η = (μ−E_c)/k_BT = {eta:.2f}). The Fermi sphere is "
               "well defined, so $k_F$, $v_F$ and $T_F$ are physically meaningful.")
elif eta < -2:
    st.warning(f"**Non-degenerate** (η = {eta:.2f}). μ lies below the band edge and "
               "Boltzmann statistics apply. There is no Fermi surface, so $k_F$, "
               "$v_F$ and $T_F$ above are formal values, not physical ones.")
else:
    st.info(f"**Intermediate** (η = {eta:.2f}). Neither limit applies cleanly; the "
            "full Fermi--Dirac integral is required, which is what is solved here.")

# ---- bands and occupation ----

left, right = st.columns(2)
E = np.linspace(-span * f.EV, Ec + span * f.EV, 2000)

with left:
    st.subheader("Bands, Fermi level and occupation")
    fig, (a1, a2) = plt.subplots(1, 2, figsize=(7.2, 4.2), sharey=True,
                                 gridspec_kw=dict(width_ratios=[1, 1.3]))
    a1.axhspan(ev(Ec), ev(E[-1]), color=style.ACOUSTIC, alpha=0.18, lw=0)
    a1.axhspan(ev(E[0]), 0.0, color=style.OPTICAL[0], alpha=0.18, lw=0)
    a1.axhline(ev(Ec), color=style.ACOUSTIC, lw=2)
    a1.axhline(0.0, color=style.OPTICAL[0], lw=2)
    a1.axhline(ev(mu), color=style.EXACT, lw=1.8, ls="--")
    a1.text(0.12, ev(Ec) + 0.04, "$E_c$", fontsize=9, color=style.ACOUSTIC)
    a1.text(0.12, -0.1, "$E_v$", fontsize=9, color=style.OPTICAL[0])
    # mu can sit within a few meV of Ec, so label it on the opposite side
    a1.text(0.78, ev(mu) - 0.085, r"$\mu$", fontsize=10)
    a1.set_xticks([])
    a1.set_ylabel("Energy (eV)")

    a2.plot(f.fermi_dirac(E, mu, T), ev(E), color=style.ACOUSTIC, lw=2)
    a2.axhline(ev(mu), color=style.EXACT, lw=1.0, ls="--")
    a2.set_xlabel("$f(E)$")
    a2.set_xlim(-0.03, 1.03)
    st.pyplot(fig)
    plt.close(fig)

with right:
    st.subheader("Density of states and occupied DOS")
    lo = min(Ec - 0.02 * f.EV, mu - 4 * f.K_B * T)
    hi = max(Ec + 0.3 * f.EV, mu + 8 * f.K_B * T)
    Ed = np.linspace(lo, hi, 2500)
    D = f.dos(Ed, m_eff, Ec) * f.EV * 1e-6
    occ = D * f.fermi_dirac(Ed, mu, T)

    fig, ax = plt.subplots(figsize=(7.2, 4.2))
    ax.semilogy(ev(Ed), np.maximum(D, 1e10), color=style.ACOUSTIC, lw=1.8,
                label="$D(E)$")
    ax.semilogy(ev(Ed), np.maximum(occ, 1e10), color=style.OPTICAL[0], lw=1.8,
                label="$D(E)f(E)$")
    ax.fill_between(ev(Ed), np.maximum(occ, 1e10), 1e10, color=style.OPTICAL[0],
```

```
                    alpha=0.25, lw=0)
    ax.axvline(ev(mu), color=style.EXACT, ls="--", lw=1.2)
    ax.set_ylim(1e12, 1e23)
    ax.set_xlabel("Energy (eV)")
    ax.set_ylabel(r"states eV$^{-1}$ cm$^{-3}$")
    ax.legend(loc="lower left", fontsize=8)
    st.pyplot(fig)
    plt.close(fig)
    st.caption("Area under $D(E)f(E)$ is the carrier concentration $n$.")

# ---- sweeps ----

st.subheader("Fermi level versus temperature and carrier concentration")
s1, s2 = st.columns(2)

with s1:
    Ts = np.linspace(5.0, 1000.0, 90)
    mus = np.array([f.solve_mu(n, t, m_eff, Ec) for t in Ts])
    fig, ax = plt.subplots(figsize=(5.6, 3.4))
    ax.plot(Ts, ev(mus - Ec), color=style.ACOUSTIC, lw=2)
    ax.axhline(0.0, color="0.55", lw=1.0)
    ax.axhline(ev(q["E_F"]), color=style.EXACT, ls=":", lw=1.0)
    ax.plot([T], [ev(mu - Ec)], marker="o", ms=7, color=style.OPTICAL[0], zorder=4)
    ax.set_xlabel("Temperature (K)")
    ax.set_ylabel(r"$\mu-E_c$ (eV)")
    ax.set_title("dotted: $E_F=\\mu(0)$;  marker: current $T$", fontsize=9)
    st.pyplot(fig)
    plt.close(fig)

with s2:
    ns = np.logspace(12, 21, 90)
    mun = np.array([f.solve_mu(x * CM3, T, m_eff, Ec) for x in ns])
    fig, ax = plt.subplots(figsize=(5.6, 3.4))
    ax.semilogx(ns, ev(mun - Ec), color=style.ACOUSTIC, lw=2)
    ax.axhline(0.0, color="0.55", lw=1.0)
    ax.plot([10.0 ** log_n], [ev(mu - Ec)], marker="o", ms=7,
            color=style.OPTICAL[0], zorder=4)
    ax.set_xlabel(r"$n$ (cm$^{-3}$)")
    ax.set_ylabel(r"$\mu-E_c$ (eV)")
    ax.set_title(r"above $E_c$ = degenerate;  marker: current $n$", fontsize=9)
    st.pyplot(fig)
    plt.close(fig)

st.caption("Both sweeps are recomputed from the full Fermi--Dirac integral at every "
           "point; neither uses the Boltzmann approximation.")
```


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